

T3 final-EFT one-loop RGE term comparison: EFT_2_after_S1_S2

All T3 particles have been integrated out at this stage. The active theory is the SM plus the dimension-five Weinberg operator. Rows are the UV model configurations that produced the matched coefficient, while columns align the additive structures in its one-loop beta function. Each populated cell contains the complete term, including its coefficient and the factor C_5 . We use $16\pi^2\mu\,dC_5/d\mu=\beta_{C_5}^{(1)}$.

$\beta_{C_5}^{(1)}$

$16\pi^2\mu\frac{dC_5}{d\mu}=\beta_{C_5}^{(1)}$

Terms 1 of 2

Column Representative term structure	
T1	$C_5g_2^2$
T2	$C_5\textit{lambda}H$
T3	$C_5ye\overline{y}e$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-3C_5g_2^2$	$2C_5\textit{lambda}H$	$-C_5ye\overline{y}e$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-3C_5g_2^2$	$2C_5\textit{lambda}H$	$-C_5ye\overline{y}e$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-3C_5g_2^2$	$2C_5\textit{lambda}H$	$-C_5ye\overline{y}e$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-3C_5g_2^2$	$2C_5\textit{lambda}H$	$-C_5ye\overline{y}e$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-3C_5g_2^2$	$2C_5\textit{lambda}H$	$-C_5ye\overline{y}e$

Terms 2 of 2

Column	Representative term structure
T1	$C_5 y d \overline{y d}$
T2	$C_5 y u \overline{y u}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$6C_5 y d \overline{y d}$	$6C_5 y u \overline{y u}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$6C_5 y d \overline{y d}$	$6C_5 y u \overline{y u}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$6C_5 y d \overline{y d}$	$6C_5 y u \overline{y u}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$6C_5 y d \overline{y d}$	$6C_5 y u \overline{y u}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$6C_5 y d \overline{y d}$	$6C_5 y u \overline{y u}$